INCH-POUND

MIL-PRF-32815
29 June 2026

PERFORMANCE SPECIFICATION

COPPER-NICKEL (70-30), LASER POWDER BED FUSION, ADDITIVELY MANUFACTURED

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers material made from additively manufactured (AM) copper-nickel (CuNi) (70-30) using the laser powder bed fusion (LPBF) process.

1.2 Classification. Material is of the following grades, as specified (see 6.2):

1.2.1 Grades. (see 4.9)

a. Grade A – Material manufactured in compliance with S9074-A2-GIB-010/AM-PBF.

b. Grade B – Material manufactured in compliance with manufacturer specifications or industry standards (see 6.3).

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-32814 - Powder Feedstock, Copper-Nickel (70-30), for use in Laser Powder Bed Fusion, Additive Manufacturing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-792 - Identification Marking Requirements for Special Purpose Components

(Copies of these publications are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-A2-GIB-010/AM-PBF - Requirements for Metal Powder Bed Fusion Additive Manufacturing

(Copies of this publication are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). This publication is/are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E3 - Standard Guide for Preparation of Metallographic Specimens

ASTM E8/E8M - Standard Test Methods for Tension Testing of Metallic Materials

ASTM E478 - Standard Test Methods for Chemical Analysis of Copper Alloys

ASTM E562 - Standard Test Method for Determining Volume Fraction by Systematic Manual Point Count

ASTM E1245 - Standard Practice for Determining the Inclusion or Second-Phase Constituent Content of Metals by Automatic Image Analysis

ASTM E1941 - Standard Test Method for Determination of Carbon in Refractory and Reactive Metals and Their Alloys by Combustion Analysis

ASTM E2465 - Standard Test Method for Analysis of Ni-Base Alloys by Wavelength Dispersive X-Ray Fluorescence Spectrometry

ASTM E2575 - Standard Test Method for Determination of Oxygen in Copper and Copper Alloys by Inert Gas Fusion

ASTM F3637 - Standard Guide for Additive Manufacturing of Metal—Finished Part Properties—Methods for Relative Density Measurement

(Copies of these publications are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS2750 - Pyrometry

(Copies of this publication are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Feedstock. Powder used as a feedstock shall be in accordance with MIL-PRF-32814.

3.2 Chemical composition. Chemical composition of the fabricated material shall conform to the percentages by weight specified in table I. Oxygen, tellurium, thallium, and selenium shall be measured and reported for information.

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TABLE I. Chemical composition (weight percent). 1/

	Minimum	Maximum
Copper	Remainder	
Lead	---	0.01
Iron	0.75	1.5
Phosphorus	---	0.02
Nickel plus cobalt	28.0	32.0
Carbon	---	0.15
Manganese	---	1.5
Sulfur	---	0.02
Silicon	---	0.5
Niobium	0.65	1.5

NOTES:
1/ Oxygen, tellurium, thallium, and selenium shall be reported for information only.

3.3 Heat treatment. Unless otherwise specified (see 6.2), material shall not be heat treated. When heat treatment is specified (see 6.2), material shall be heat treated under vacuum or an inert argon atmosphere and in furnaces in accordance with the pyrometry requirements of SAE AMS2750, class 5 or better.

3.4 Tensile properties. The tensile properties of all material shall meet or exceed the minimum values specified in table II. Reduction of area shall be measured and reported for information.

TABLE II. Tensile properties.

Property	Minimum value
Ultimate tensile strength (ksi)	60
Yield strength, 0.2% offset (ksi)	40
Elongation in 2 inches (%)	25
Reduction of area (%)	Information only

3.5 Relative density. Unless otherwise specified (see 6.2), the relative density of the material shall be not less than 99.8 percent.

3.6 Weld repair. Unless otherwise specified (see 6.2), weld repair shall be prohibited.

3.7 Additional mechanical properties. Additional mechanical properties shall be as specified (see 6.2).

3.8 Detrimental material controls. During the manufacturing processes, examinations, and tests, the material shall not come into direct contact with mercury or any of its compounds, nor with any mercury-containing device employing a single boundary of containment. Additional detrimental material controls shall be as specified (see 6.2).

3.9 Identification marking. Material other than specimens used for conformance inspections shall be marked with the number and revision of this specification, manufacturer name or CAGE code, and manufacturing lot number. Markings shall be permanently applied in accordance with MIL-STD-792; an alternate marking method is permissible, provided it is an available option in the contractually invoked drawing or specification. Markings shall

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be located so as not to affect the form, fit, or function of the material and must be legible after all required surface finishing and post-processing operations.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.2).

4.2 Conformance inspection. Conformance inspection shall include the examinations and tests of table III.

TABLE III. Conformance inspections.

Examinations and tests	Requirement	Conformance inspection	Sampling requirements
Powder feedstock	3.1	4.9	---
Chemical composition	3.2	4.5	4.4.1 and 4.4.4
Heat treatment	3.3	4.9	---
Tensile properties	3.4	4.6 and 4.9	4.4.2, 4.4.4, and 4.4.5
Relative density	3.5	4.7 and 4.9	4.4.3, 4.4.4, and 4.4.5
Weld repair	3.6	4.9	---
Additional mechanical properties	3.7	4.8 and 4.9	---
Detrimental material controls	3.8	4.9	---
Identification marking	3.9	4.9	---

4.3 Lot size. Unless otherwise specified (see 6.2), material produced in one build cycle (see 6.11.1) shall constitute a manufacturing lot. When heat treatment is required (see 3.3), material produced in one build cycle and the same furnace load shall constitute a manufacturing lot.

4.4 Sampling.

4.4.1 Chemical composition. A minimum of one sample shall be tested from each lot in its final heat treated condition.

4.4.2 Tensile properties. A minimum of two tensile specimens shall be tested from each lot in its final heat treated condition. Sample placement on the build plate shall be representative of the lowest tensile performance region of the laser being sampled, as identified through prior characterization, so that specimen location does not bias mechanical test results in a non-conservative direction (see 6.5). Tensile specimens shall be machined from bulk deposition or near-net shape blanks. Tensile specimens shall conform to ASTM E8/E8M standard round specimen size 3 or larger with a gage length four times the diameter. At least one specimen shall be oriented with its length in the Z-axis and at least one shall be oriented with its length in the XY-plane (see 6.11.3 for orientation definitions). See 4.4.5 for additional sampling requirements for multi-laser LPBF systems. See 6.6 for recommendations regarding interrogating build cycle interruptions. See 6.7 for recommendations regarding specimen quantity for high criticality applications.

4.4.3 Relative density. A minimum of one representative sample shall be tested from each lot for analysis. Sample placement on the build plate shall be representative of the lowest relative density performance region of the laser being sampled, as identified through prior characterization, so that specimen location does not bias relative density results in a non-conservative direction (see 6.5). The cross-section for relative density measurement shall be taken in the XZ or YZ plane. See 6.11.3 for definition of LPBF build orientations and planes. The sample or samples shall include at least one external surface in addition to interior material. Sample dimensions shall be such

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that metallographic imaging covers a continuous minimum sample area of 0.25 square inch. The relative density sample shall be tested in the final heat treated condition. See 4.4.5 for required additional sampling of multi-laser LPBF systems.

4.4.4 Resampling and retesting. Test results that do not meet this specification may be addressed by testing two additional samples taken from the same lot as the original sample. If the initial sample fails, the two re-test samples shall both meet the minimum requirements, and the average of all three samples (initial and two re-tests) shall meet the minimum requirements. If either of the two re-test samples fails, the lot shall be rejected. If any two samples of any type of test fail to meet the acceptance criteria, then the lot shall be rejected. When specified (see 6.2), any lot may be rejected based on a single tensile or relative density test.

4.4.5 Sampling of multi-laser systems. Conformance to the requirements for tensile properties and relative density shall be demonstrated for each laser used in production. To reduce testing burden, more than one laser may optionally be used to produce any given conformance inspection sample. To demonstrate conformance for any individual laser, at a minimum, one fifth (20 percent) of the tested volume (e.g., gauge region of a horizontal tensile bar or sample area inspected for relative density) of the conformance sample shall have been processed by that laser. For Z-axis oriented tensile bars, each laser under evaluation shall produce a minimum build height equal to 1.2 times the nominal specimen diameter. See 6.8 for examples.

4.5 Chemical composition analysis. The chemical composition of each lot shall be tested and analyzed by accepted analytical methods. In case of disagreements, the referee methods shall be ASTM E2575 for oxygen, ASTM E1941 for carbon, ASTM E2465 for niobium, and ASTM E478 for the remaining compositional requirements. The chemical composition of each lot shall conform to the compositions shown in table I.

4.6 Tensile testing. The tensile specimens shall be tested in accordance with ASTM E8/E8M and shall meet the requirements of table II (see 3.4).

4.7 Relative density. The relative density shall be measured according to ASTM F3637 metallography guidance. Metallography shall be performed in accordance with ASTM E3. Image acquisition shall utilize a method conforming to ASTM E562 or ASTM E1245, with a minimum magnification of 10X; higher magnifications are permitted. Support structures shall be excluded from imaging and measurement. Images shall be processed to a grayscale or binary image and pixels counted using a histogram in accordance with ASTM E1245. See 6.9 for reporting recommendations.

4.8 Additional mechanical properties. Test methods, sampling requirements, and acceptance criteria for testing of additional mechanical properties shall be as specified (see 6.2).

4.9 Certificate of conformance. Unless otherwise specified (see 6.2), a certificate of conformance (COC) shall be provided for each lot of material offered for acceptance in accordance with the lot definition specified in 4.3. The COC shall include the following:

a. Statement that each lot has been sampled, tested, and inspected in accordance with the requirements specified herein.

b. For grade A material (see 1.2.1), record of approval from the Technical Authority (see 6.11.4) or authorized representative that indicates the manufacturer is in compliance with S9074-A2-GIB-010/AM-PBF.

c. For grade B material (see 1.2.1), statement specifying the manufacturer or industry LPBF process standard(s) (see 6.3) used to determine material compliance and record of approval from the Technical Authority (see 6.11.4) or authorized representative thereof.

d. Statement that powder feedstock is in accordance with the requirements specified in MIL-PRF-32814 (see 3.1).

e. If heat treatment is required (see 3.3), complete record of all heat treatments, including time, temperatures, atmosphere, and heating and cooling rates.

f. If weld repair is permitted (see 3.6), complete record of all repaired defects with location, weld repair inspection results, post-weld heat treatment (if any), and associated weld procedure approvals.

g. Statement certifying that detrimental material control requirements have been met (see 3.8).

h. Statement certifying that all material markings comply with the requirements of this specification (see 3.9).

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i. Complete record of testing (see table III) used to demonstrate compliance with this specification, including any failed tests.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. LPBF CuNi (70-30) AM material is intended for replacement of castings specified in ASTM B369/B369M, "Standard Specification for Copper-Nickel Alloy Castings," for use in Navy applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grade required (see 1.2).

c. When heat treatment is required (see 3.3).

d. Relative density if other than specified (see 3.5).

e. When weld repair is permitted (see 3.6).

f. Additional mechanical properties (see 3.7).

g. When needed, detrimental material control requirements (see 3.8).

h. Variable lot size when required (see 4.3).

i. When tests that do not meet this specification are not allowed to be repeated by additional samples taken from the original lot (see 4.4.4).

j. Test specimen numbers, geometry, orientation, and test method if other than specified (see 4.8).

k. When a certificate of conformance is not required (see 4.9).

l. Packaging requirements (see 5.1 and 6.10).

m. Surface finish requirements (see 6.4).

n. Additional sampling for build cycle interruptions, when required (see 6.6).

o. Requirements for increased sampling (see 6.7).

6.3 LPBF qualification. Grade B material requires the disclosure of a process qualification standard. Appropriate industry standards include SAE AMS7003, "Laser Powder Bed Fusion Process," AWS D20.1/D20.1M, "Specification for Fabrication of Metal Components Using Additive Manufacturing," or other standards that provide a framework to prove that the LPBF process is capable of producing material of consistent quality.

6.4 Surface finish. Material produced using LBPF has a rough surface that may be associated with a reduction in corrosion resistance and fatigue life. A rough surface may also reduce inspectability by restricting the use of NDT methods such as ultrasonic testing and dye penetrant testing, particularly when average roughness exceeds 250 microinch. Material is often post-processed to achieve the required surface finish. Prescriptive surface finishing requirements should be incorporated if necessary (see 6.2).

6.5 Locating the regions of lowest performance. The tensile properties and relative density of samples placed in differing locations on the build platform (see 6.11.2) may differ significantly due to changes in the local process conditions. For example, local changes in the gas flow and laser spot size can affect relative density levels, which in turn can affect tensile properties. Regional variations in tensile properties and relative density are typically established during machine qualification (see 6.3). If the region of lowest performance has not been determined, it

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may be assumed to be the region accessed by the relevant laser that is downstream of the gas flow and toward the end of the recoating pass.

6.6 Specimen for interrogating build cycle interruptions. Interruptions to the build cycle may occur for a variety of reasons, both planned and unplanned. Build interruptions may result in a lack of fusion at the interrupted layer. A vertical tensile coupon may be extended through the height of the build to assess the properties of the material at the interrupted layer in the case of such an event. Build cycle interruption specimens should be as specified (see 6.2).

6.7 Additional specimens. Specimen quantity and test type may be increased for a component of high criticality. For instance, additional tensile tests, or other tests including fatigue are encouraged for high criticality parts, when those properties are relevant to the application. For additional tests, the method, geometry (to include surface condition), orientation, and acceptance criteria should be specified. Increased sampling should be as specified (see 6.2).

6.8 Conformance testing for multi-laser systems. This discussion gives an example of how conformance for mechanical properties and relative density may be shown for a system that uses more than one laser. Note that chemical composition is to be measured per lot, not per laser, and so is not included in this discussion. Conformance is only required for those lasers used to produce material under this specification. The following example uses a four-laser system, but the discussion applies equally to systems with any number of lasers:

a. In the case of regional lasers (lasers confined to a specific corner or region of the build platform area), the producer may use the left two lasers to produce one set of conformance samples and the right two lasers to produce a second set in their respective overlap regions. Optionally, the front and back lasers may be similarly combined in their overlap region.

b. In the case of global lasers (lasers that span and are commonly used across the whole or majority of the build platform area), a single set of conformance samples may be produced. These samples could be located anywhere on the build platform. See 6.5 for location recommendations.

c. If the producer opts not to use multiple lasers to produce conformance samples, each individual laser is required to produce a complete set of tensile and relative density conformance samples.

d. In all cases where producers use more than one laser to produce a conformance sample, the contribution of each laser used in production is required to meet the 20 percent threshold (see 4.4.5). Conformance to this requirement may be stated via the COC. For example, in the case of the four-laser machine where all four lasers are producing one set of conformance samples, each laser should be shown to have produced at minimum one fifth of the gage section of each sample.

e. The requirements for Z-axis oriented tensile specimens differ from horizontal tensile specimens and relative density specimens, in that for Z-axis oriented tensile bars, each laser under evaluation should produce a minimum build height equal to 1.2 times the nominal specimen diameter. For example, if the producer elects to use a tensile sample with 0.25-inch gage diameter, each laser should thus produce at least 0.3 inch of the length of the gage section of the tensile bar, in order to properly assess the performance of that laser.

6.9 Relative density reporting. It is recommended that the processed image(s) and captured image(s) be reported together. If multiple images are analyzed, a stereological confidence interval may be computed from the images in accordance with ASTM E562 or ASTM E1245. The lower limit of the confidence interval will serve as the relative density measurement.

6.10 Packaging marking. In addition to other packaging and labeling requirements required by the contract (see 6.2), each unit, intermediate, and exterior shipping container should be marked with the material specification number, manufacturer's name or CAGE code, and manufacturing lot number. This marking should be placed on each MIL-STD-129, "Military Marking for Shipment and Storage," label required by the contract and should be placed either below the identification test marking or in a conspicuous location on the identification marked side of the container.

6.11 Definitions.

6.11.1 Build cycle. See ISO/ASTM 52900, "Additive manufacturing — General principles — Fundamentals and vocabulary," for further clarification.

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6.11.2 Build platform. See ISO/ASTM 52900, "Additive manufacturing — General principles — Fundamentals and vocabulary," for further clarification.

6.11.3 LPBF build orientations. Figure 1 is a schematic representation of the orientation of planes in the LPBF process. The X-axis lies in the plane of the build platform and is parallel to the front of the machine. The Y-axis lies in the plane of the build platform and is perpendicular to the X-axis. The Z-axis is perpendicular to the X-axis and Y-axis and is normal to the layers. The XZ plane is a plane containing the X-axis and Z-axis. The YZ plane is a plane containing the Y-axis and Z-axis. See ISO/ASTM 52921, "Standard Terminology for Additive Manufacturing — Coordinate Systems and Test Methodologies," for further clarification.

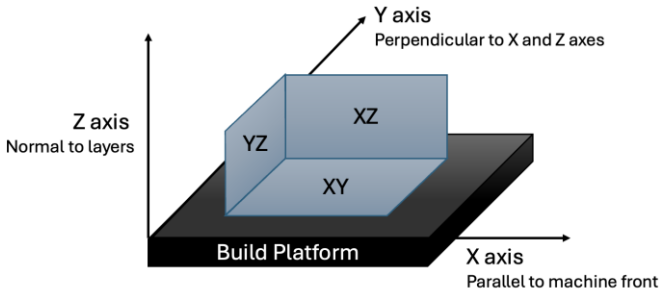


Figure 1. Schematic showing LPBF XY, YZ, and XY planes.

6.11.4 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05T may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.12 Subject term (key word) listing.

CuNi (70-30)
3-D Printing
Casting Substitute
Metal Powder Printing
Metal Printing
Powdered Metal

CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – SH

Preparing activity:
Navy – SH
(Project AMPR-2026-002)

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